
LatentGoalOps: Supplementary Material

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1 Protocol Snapshot

2 This supplement is intentionally narrower than the main paper. It records the frozen proto-
 3 col, serving path, supplementary diagnostics, and provenance for the strengthened bundle at
 4 outputs/paper-freeze-2026-04-18. It does not repeat the main model table or baseline ta-
 5 ble, which are already reported in the main paper. The paper-facing benchmark has three sequential
 6 environments; the main manuscript centers P1 and P2 as the result-bearing anchor tasks and treats
 7 P3 as shorter breadth evidence. Auxiliary one-shot tasks in the code release are for smoke tests and
 8 extensibility, not headline claims.

Paper ID	Environment	Internal id	Horizon	Purpose
P1	Startup Week	task3	10	Long-horizon latent-goal tracking under delayed ef- fects.
P2	Incident Response Week	task6	7	Shift adaptation under governance and recovery pres- sure.
P3	Quarterly Headcount Plan	task7	4	Shorter planning-breadth task under reprioritization.

Table 1: Paper-facing environments in the frozen benchmark. Internal ids are kept only for code and manifest provenance.

9 2 Latent Objective and Public Schema

10 The hidden objective is a structured latent state rather than a single class label. It combines an
 11 archetype, a four-channel weight vector, a primary KPI, and discrete operating attributes such as
 12 risk posture, planning horizon, segment focus, and governance strictness. Table 2 gives the base
 13 archetypes used in the frozen benchmark.

Archetype	Channel weights	KPI	Operating attributes
Growth	(0.55, 0.15, 0.20, 0.10)	DAU	aggressive risk; quarterly planning; flexible gover- nance; self-serve segment
Retention	(0.10, 0.60, 0.15, 0.15)	D30 retention	balanced risk; quarterly planning; strict gover- nance; enterprise segment
Revenue	(0.10, 0.20, 0.55, 0.15)	MRR	balanced risk; quarterly planning; moderate gover- nance; mid-market segment
Efficiency	(0.10, 0.15, 0.15, 0.60)	Ops margin	conservative risk; strategic planning; strict gover- nance; enterprise segment

Table 2: Base latent-goal archetypes. Channel order is growth, retention, revenue, efficiency.

14 In the frozen configuration, shifts occur with probability 0.6; shift steps are sampled from $\{3, 4, 5\}$;
 15 and the shift can be abrupt or drift-like. The public schema exposes recurring customer accounts,
 16 stakeholders, internal teams, market context, and governance constraints, while redacting the latent
 17 channel decomposition and hidden utility weights.

18 3 Formal Benchmark Definition

19 Let x_t denote the hidden environment state at step t , let $o_t = \psi(x_t)$ be the public observation
 20 exposed to the model, and let $a_t \in \mathcal{A}$ be a structured action emitted through the benchmark’s action
 21 schema. An episode is a partially observed trajectory

$$\tau = (o_0, a_0, o_1, a_1, \dots, o_T).$$

22 The hidden objective state is g_t , which determines which latent business outcomes matter at time t .
 23 The agent never observes g_t directly.

24 **Latent utility.** The environment maintains a normalized four-channel metric vector $m_t \in [0, 1]^4$
 25 over growth, retention, revenue, and efficiency. The active latent objective induces a weight vector
 26 $w_t \in \Delta^4$. The latent utility used by the graders is

$$U_t = \text{clip}_{[0,1]} \left(\sum_{c \in \mathcal{C}} m_{t,c} w_{t,c} \right), \quad \mathcal{C} = \{\text{growth, retention, revenue, efficiency}\}.$$

Symbol	Meaning
x_t	hidden environment state at step t
o_t	public observation exposed to the model
a_t	structured action emitted by the model
g_t	active hidden objective state
m_t	normalized four-channel public metric vector
w_t	active latent utility weight vector
U_t	latent utility under the active objective
q_t	normalized decision quality at step t
$\tilde{q}_t$	stale-objective decision quality after a shift
A	adaptation score
C	trajectory coherence score
K	constraint-satisfaction score
Δ_{shell}	native-only minus overall score
Δ_{shift}	unshifted minus shifted score

Table 3: Core notation used in the formal benchmark definition.

Task	Utility	Adaptation	Coherence	Constraints
Startup Week	0.5000	0.2500	0.1500	0.1000
Incident Response Week	0.4634	0.2927	0.1220	0.1220
Quarterly Headcount Plan	0.6364	0.2614	0.0568	0.0454

Table 4: Prespecified score coefficients for the paper-facing sequential tasks. These weights make adaptation and constraint behavior visible while preserving final latent utility as the largest single component.

When a hidden shift is present, the active objective changes from a base objective $g^{(0)}$ to a shifted objective $g^{(1)}$ at step t_* . The implementation supports abrupt shifts and drift shifts. If the shift is abrupt, $w_t = w^{(1)}$ for all $t \geq t_*$. If the shift is a drift over duration d , then the active weight vector is a normalized convex blend

$$w_t = \text{Normalize}\left((1 - \rho_t)w^{(0)} + \rho_t w^{(1)}\right), \quad \rho_t = \text{clip}_{[0,1]}\left(\frac{t - t_* + 1}{d}\right).$$

The shift is never explicitly announced to the model; it must be inferred from changed consequences and public operating evidence.

For the three paper-facing environments, the overall score is a weighted combination of final latent utility U , adaptation A , coherence C , and constraint satisfaction K :

$$S_{P1} = 0.50 U + 0.25 A + 0.15 C + 0.10 K,$$

$$S_{P2} = 0.4634 U + 0.2927 A + 0.1220 C + 0.1220 K,$$

$$S_{P3} = 0.6364 U + 0.2614 A + 0.0568 C + 0.0454 K.$$

Let $q_t \in [0, 1]$ be the normalized decision quality at step t , defined as agent decision value divided by oracle decision value under the *active* objective. After a shift, the implementation also computes a stale-objective quality $\tilde{q}_t$ whenever the base objective remains well-defined. The first detected adaptation step for P1 and P2 is

$$t_{\text{det}} = \min \{t \geq t_* : q_t \geq \max(0.72, \tilde{q}_t + 0.05)\},$$

with the fallback rule $t_{\text{det}} = T$ if no post-shift step crosses the threshold. The remaining adaptation terms are

$$R = \frac{1}{|D|} \sum_{t \in D} q_t, \quad \text{speed} = 1 - \frac{t_{\text{det}} - t_*}{\max(T - t_* + 1, 1)},$$

where $D = \{t_{\text{det}}, \dots, T\}$ if detection occurs and $D = \emptyset$ otherwise. For P1 and P2, the adaptation score is

$$A = 0.55 \cdot \text{speed} + 0.45 \cdot R.$$

P3 uses the same final weighted-score framework but a shorter-horizon adaptation rule. Its detected adaptation step uses threshold

$$q_t \geq \max(0.60, \tilde{q}_t + 0.02),$$

Task	Detection rule	Adaptation mix	Coherence and no-shift handling
P1 Startup Week	$q_t \geq \max(0.72, \bar{q}_t + 0.05)$	0.55 speed, 0.45 recovery	Coherence ignores the true pivot; unshifted episodes omit adaptation and renormalize.
P2 Incident Response Week	$q_t \geq \max(0.72, \bar{q}_t + 0.05)$	0.55 speed, 0.45 recovery	Coherence ignores the true pivot; unshifted episodes omit adaptation and renormalize.
P3 Quarterly Headcount Plan	$q_t \geq \max(0.60, \bar{q}_t + 0.02)$	0.40 speed, 0.60 recovery	Adaptation is multiplied by post-shift coherence; unshifted episodes omit adaptation and renormalize.

Table 5: Adaptation parameters for P1–P3. Shifted episodes without detection receive zero post-detection recovery; unshifted episodes omit adaptation and renormalize the score.

with the same fallback of no post-detection recovery when the threshold is not crossed. Let R_7 be mean post-detection decision quality, let speed_7 be the normalized detection-speed term over the post-shift window, and let C_{post} be trajectory coherence over post-shift actions. Then

$$A_7 = (0.40 \cdot \text{speed}_7 + 0.60 \cdot R_7) C_{\text{post}}.$$

When an episode has no hidden shift, the adaptation component is marked not applicable and the weighted score is renormalized over the active components only. When a shift exists but no post-shift decision row is available, adaptation is scored as 0.0.

Trajectory coherence is the mean within-regime cosine similarity between successive action-strategy embeddings, ignoring the single pivot induced by a true objective shift. To separate latent-policy failure from interface failure, the paper also reports

$$\Delta_{\text{shell}} = S_{\text{native}} - S_{\text{overall}}, \quad \Delta_{\text{shift}} = S_{\text{unshifted}} - S_{\text{shifted}},$$

where S_{native} is the mean score over episodes that remain fully native to the structured action interface and S_{overall} includes fallback-affected episodes.

4 Serving, Metadata, and Interface

The strengthened paper protocol uses runtime version 2026-04-07-shell-overhaul-v6 and freeze date 2026-04-18. The resolved manifest is the source of truth for the freeze: the analysis root collects validated model runs, baseline runs, diagnostics, and targeted top-ups under one dated paper bundle while preserving their original relative run directories. The artifact bundle contains 10 locally served primary benchmark configurations and one API-served external anchor across four served families:

- Qwen3.5: 2B, 9B, 27B [Team, 2025b];
- DeepSeek-R1: 1.5B, 8B, 14B [DeepSeek-AI et al., 2025];
- Gemma4: 2B, 4B, 26B [Team et al., 2024, Team, 2025a];
- GPT-OSS: 20B locally, 120B as an external API anchor [OpenAI, 2025].

The strengthened bundle contains:

- 630 primary model episode rows;
- 60 external-anchor episode rows;
- 360 baseline episode rows;
- 84 leakage-probe rows;
- generated tables, figures, case studies, and protocol descriptors tied to the frozen bundle.

The online environment also maintains a leakage-safe shaped reward used during interaction, but not as the final benchmark metric.

The strengthened artifact bundle mixes two execution paths: local Ollama serving for the 10 primary benchmark configurations and OpenRouter API serving for GPT-OSS-120B. The provider difference is made explicit and GPT-OSS-120B is reported only as an external anchor rather than as a load-bearing point in the main scaling matrix.

Model	Split	Provider	In tok (K)	Out tok (K)	Cost (\$)	Native	Empty fb
DeepSeek 1.5B	core	Ollama local	772.2	5.0	0.000	1.00	0.00
DeepSeek 1.5B	heldout	Ollama local	786.7	4.8	0.000	1.00	0.00
DeepSeek 8B	core	Ollama local	714.4	5.2	0.000	1.00	0.00
DeepSeek 8B	heldout	Ollama local	717.8	5.4	0.000	1.00	0.00
DeepSeek 14B	core	Ollama local	724.9	5.1	0.000	1.00	0.00
DeepSeek 14B	heldout	Ollama local	706.2	5.3	0.000	1.00	0.00
Gemma 2B	core	Ollama local	822.0	6.1	0.000	1.00	0.00
Gemma 2B	heldout	Ollama local	825.1	6.0	0.000	1.00	0.00
Gemma 4B	core	Ollama local	752.6	5.6	0.000	1.00	0.00
Gemma 4B	heldout	Ollama local	747.2	5.6	0.000	1.00	0.00
Gemma 26B	core	Ollama local	749.3	42.8	0.000	0.70	0.30
Gemma 26B	heldout	Ollama local	748.2	38.8	0.000	0.73	0.27
GPT-OSS 20B	core	Ollama local	812.5	16.8	0.000	1.00	0.00
GPT-OSS 20B	heldout	Ollama local	784.2	8.6	0.000	1.00	0.00
Qwen 2B	core	Ollama local	779.4	5.1	0.000	1.00	0.00
Qwen 2B	heldout	Ollama local	785.1	5.0	0.000	1.00	0.00
Qwen 9B	core	Ollama local	704.2	5.2	0.000	1.00	0.00
Qwen 9B	heldout	Ollama local	694.2	5.8	0.000	1.00	0.00
Qwen 27B	core	Ollama local	1408.3	10.4	0.000	0.97	0.03
Qwen 27B	heldout	Ollama local	430.1	196.9	0.000	0.47	0.53

Table 6: Provider/runtime summary for the frozen primary-task aggregate. This table is included because benchmark interpretation depends on serving path, token pressure, and interface reliability.

81 The runtime table reports the visible serving path, token volume, cost, and interface-reliability summaries for each primary model/split. Exact raw model aliases, run roots, and role labels are preserved
82 in `metadata/paper_protocol_resolved.json` and `metadata/run_manifest.json`. The release also includes provider-path and token-count records in `paper/release/COMPUTE.md`; the
83 original worker inventory is partially preserved, so the checklist marks full compute inventory disclosure conservatively while keeping the runnable evaluation artifact reproducible from the shipped
84 manifest.

88 5 Action Interface and Failure Taxonomy

89 The action interface is intentionally narrow. On P1 and P2, a valid action contains a task id
90 plus a JSON-like payload with four core keys: `chosen_initiatives`, `messaging_action`,
91 `pricing_change_pct`, and `support_policy`. Optional rationale, memory, and belief notes may
92 also be present. The allowed messaging values are growth push, retention campaign, revenue upsell,
93 cost comms, and none. The allowed support policies are premium SLA, balanced triage, automation
94 first, and incident swarm. P3 uses the same action envelope but replaces initiative/package selection
95 with `budget_allocations`. Validation rejects irrelevant cross-task fields.

Signal	Operational meaning	Log field	Why it matters
Native step	Action parses and satisfies the schema	strict action valid	Lets us attribute reward to the model's policy rather than fallback behavior.
Empty fallback	Parser returns an empty/default action	empty fallback	Damaging in sequential tasks because the world advances while the model does little useful work.
Cap hit	Response exceeds the output cap	response cap hit	Separates verbosity-related truncation from ordinary policy error.
Parse repair	Action required bounded repair before execution	appendix only	Diagnostic only. Tests whether failure is a lightweight parser issue or a deeper interface-control problem.
Native episode rate	Fraction of episodes that remain native throughout	aggregated metric	Primary paper signal for shell reliability at the episode level.

Table 7: Action-interface failure taxonomy used throughout the paper.

96 6 Supplementary Reference Tables

97 The main model and baseline tables are reported in the main paper. This section keeps only the
98 external anchor and the diagnostics that are specific to the supplementary analysis.

99 The sensitivity table is best read as a construct check. Baseline ordering is stable for the origi-
100 nal, utility-only, balanced, adaptation-heavy, and adaptation-light variants on both splits. The no-
101 coherence variant is included as a useful boundary case: it removes one piece of the sequential-
102 behavior signal and breaks core baseline ordering. This supports keeping coherence as a small but
103 nonzero component of the prespecified score. The utility-only rows also clarify why LatentGoalOps
104 should not be interpreted as a final-state utility benchmark alone: once adaptation is removed, the
105 myopic-oracle gap shrinks sharply even though the shifted-episode penalty remains visible.

106 The shell-controlled diagnostic is an appendix mechanism check focused on the two week-long an-
107 chor tasks, Startup Week and Incident Response Week. It is not used as a replacement for the head-
108 line three-task aggregate; instead, it tests whether parse repair alone explains the largest interface-
109 mediated drops.

110 Full per-environment and subscore tables are shipped as machine-readable CSV files and LaTeX
111 fragments under `outputs/paper-freeze-2026-04-18/tables` and `paper/latex/generated`.
112 They are intentionally not expanded here so the supplement remains readable and free of float-only
113 table pages.

Model	Split	Provider	Episodes	Overall	95% CI	Native rate	Empty fallback
GPT-OSS 120B	core	OpenRouter API	30	0.381	[0.348, 0.415]	0.60	0.40
GPT-OSS 120B	heldout	OpenRouter API	30	0.365	[0.339, 0.396]	0.43	0.57

Table 8: External anchors excluded from the primary benchmark matrix because they were not run under the same local serving path. They are retained for transparency but are not used as load-bearing within-family scaling evidence.

Check	Value	Passes	Threshold
Core baseline ordering	$0.303 < 0.346 < 0.599$	Yes	random < visible heuristic < myopic oracle
Core oracle gap	0.296	Yes	≥ 0.20
Heldout baseline ordering	$0.310 < 0.371 < 0.598$	Yes	random < visible heuristic < myopic oracle
Heldout oracle gap	0.289	Yes	≥ 0.20
Silent-shift penalty	0.172	Yes	median unshifted-minus-shifted gap ≥ 0.10

Table 9: Selected benchmark health checks used in the paper.

Variant	Split	Best primary	Best score	Oracle gap	Shift gap	Baseline order
original	core	Qwen 27B	0.443	0.156	0.154	Yes
original	heldout	DeepSeek 8B	0.446	0.152	0.183	Yes
utility-only	core	Qwen 27B	0.392	0.012	0.072	Yes
utility-only	heldout	GPT-OSS 20B	0.393	0.016	0.083	Yes
balanced	core	DeepSeek 14B	0.494	0.087	0.185	Yes
balanced	heldout	DeepSeek 8B	0.499	0.081	0.209	Yes
adaptation-heavy	core	Qwen 27B	0.463	0.089	0.214	Yes
adaptation-heavy	heldout	DeepSeek 8B	0.462	0.088	0.249	Yes
adaptation-light	core	DeepSeek 1.5B	0.519	0.057	0.109	Yes
adaptation-light	heldout	Qwen 2B	0.523	0.056	0.118	Yes
no-coherence	core	Qwen 27B	0.430	0.061	0.169	No
no-coherence	heldout	DeepSeek 8B	0.429	0.060	0.207	Yes

Table 10: Offline score-weight sensitivity from frozen final-episode grader outputs. “Shift gap” is median unshifted-minus-shifted gap across primary models. The utility-only variant is included to show that the benchmark’s main signal depends on the intended adaptation construct, not only final utility.

Task	Split	Full	Numeric	Sender	Order	Max restricted	Goal tokens
Startup Week	core	0.20	0.05	0.25	0.25	0.30	0.00
Startup Week	heldout	0.20	0.15	0.20	0.25	0.25	0.00
Incident Response Week	core	0.20	0.10	0.30	0.25	0.30	0.00
Incident Response Week	heldout	0.15	0.20	0.30	0.25	0.30	0.00
Quarterly Headcount Plan	core	0.25	0.25	0.25	0.25	0.35	0.00
Quarterly Headcount Plan	heldout	0.20	0.15	0.25	0.25	0.30	0.00

Table 11: Compact restricted-view leakage summary for paper-facing tasks. Chance is 0.25 for the four-way archetype probe. “Max restricted” is the maximum over non-full views in the frozen audit and is interpreted as a screening diagnostic rather than a proof of no shortcut.

Model	Core P1		Core P2		Heldout P1		Heldout P2	
	Adapt.	Zero rate	Adapt.	Zero rate	Adapt.	Zero rate	Adapt.	Zero rate
Qwen 2B	0.000	1.00	0.000	1.00	0.000	1.00	0.000	1.00
Qwen 9B	0.000	1.00	0.181	0.78	0.198	0.67	0.000	1.00
Qwen 27B	0.117	0.82	0.060	0.91	0.000	1.00	0.000	1.00
DeepSeek 1.5B	0.000	1.00	0.000	1.00	0.000	1.00	0.000	1.00
DeepSeek 8B	0.101	0.89	0.190	0.78	0.180	0.78	0.185	0.78
DeepSeek 14B	0.090	0.89	0.195	0.78	0.195	0.67	0.000	1.00
Gemma 2B	0.000	1.00	0.000	1.00	0.000	1.00	0.000	1.00
Gemma 4B	0.132	0.78	0.181	0.78	0.237	0.67	0.000	1.00
Gemma 26B	0.161	0.78	0.298	0.61	0.000	1.00	0.033	0.94
GPT-OSS 20B	0.098	0.78	0.277	0.67	0.057	0.89	0.000	1.00

Table 12: Supplementary adaptation audit. “Adapt.” is mean adaptation score on shifted episodes; “Zero rate” is the fraction of shifted episodes with zero adaptation score.

Model	Split	Frozen overall	Shell overall	Shell assisted	Diagnostic native	Repair rate
Gemma 26B	core	0.395	0.285	0.278	0.00	0.10
Gemma 26B	heldout	0.361	0.287	0.353	0.00	0.10
Qwen 27B	core	0.443	0.333	0.340	0.05	0.40
Qwen 27B	heldout	0.385	0.345	0.375	0.25	0.20

Table 13: Appendix shell-controlled diagnostic with parse repair enabled and heuristic rescue disabled. “Diagnostic native” is measured inside this separate diagnostic rerun condition and is not the frozen benchmark native episode rate. “Shell assisted” is the assisted aggregate reported by the diagnostic runner.

Family	Split	Transition	Δ overall	Δ native-only	Δ native rate	Diagnosis
DeepSeek-R1	core	DeepSeek 1.5B $\rightarrow$ DeepSeek 8B	-0.026	-0.026	0.00	native policy regression
DeepSeek-R1	heldout	DeepSeek 1.5B $\rightarrow$ DeepSeek 8B	0.016	0.016	0.00	gain
DeepSeek-R1	core	DeepSeek 8B $\rightarrow$ DeepSeek 14B	0.035	0.035	0.00	gain
DeepSeek-R1	heldout	DeepSeek 8B $\rightarrow$ DeepSeek 14B	-0.017	-0.017	0.00	native policy regression
Gemma4	core	Gemma 2B $\rightarrow$ Gemma 4B	0.016	0.016	0.00	gain
Gemma4	heldout	Gemma 2B $\rightarrow$ Gemma 4B	0.013	0.013	0.00	gain
Gemma4	core	Gemma 4B $\rightarrow$ Gemma 26B	-0.025	0.013	-0.30	interface regression
Gemma4	heldout	Gemma 4B $\rightarrow$ Gemma 26B	-0.067	-0.044	-0.27	interface regression
Qwen3.5	core	Qwen 2B $\rightarrow$ Qwen 9B	-0.008	-0.008	0.00	flat
Qwen3.5	heldout	Qwen 2B $\rightarrow$ Qwen 9B	-0.013	-0.013	0.00	native policy regression
Qwen3.5	core	Qwen 9B $\rightarrow$ Qwen 27B	0.021	0.022	-0.03	gain
Qwen3.5	heldout	Qwen 9B $\rightarrow$ Qwen 27B	-0.037	0.035	-0.53	interface regression

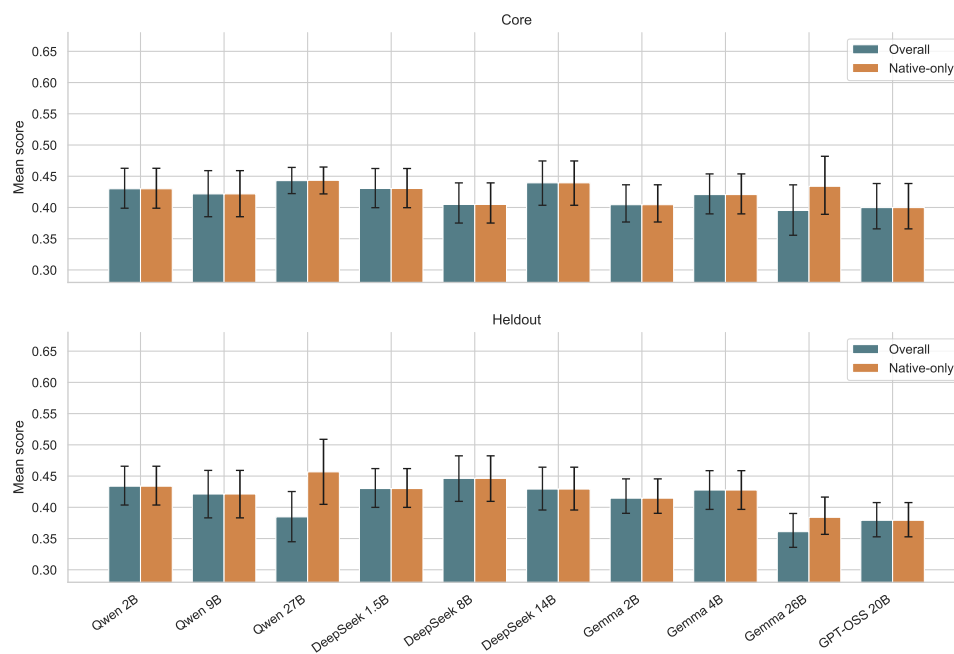
Table 14: Scaling-transition summary. Native-only deltas help separate clean policy regressions from interface-mediated regressions.

114 **7 Supplementary Figures**

115 **Leakage Audit Details**

116 The leakage figure condenses the frozen leakage table `tables/leakage_results.csv` inside the
117 paper-freeze artifact bundle. On the primary tasks, the full-observation centroid probe is 0.20
118 on core Startup Week, 0.20 on core Incident Response Week, 0.25 on core Quarterly Headcount
119 Plan, and 0.15–0.20 on the corresponding heldout slices. Restricted views such as `numeric_only`,
120 `sender_only`, and `stakeholder_order_only` remain near the four-way chance regime, while
121 `explicit_goal_token_rate` is 0.0 in every frozen probe output.

Observed score vs. native-only score (native-only estimates marked when native n is small)



Structured-action reliability remains a major failure mode in the frozen ladder

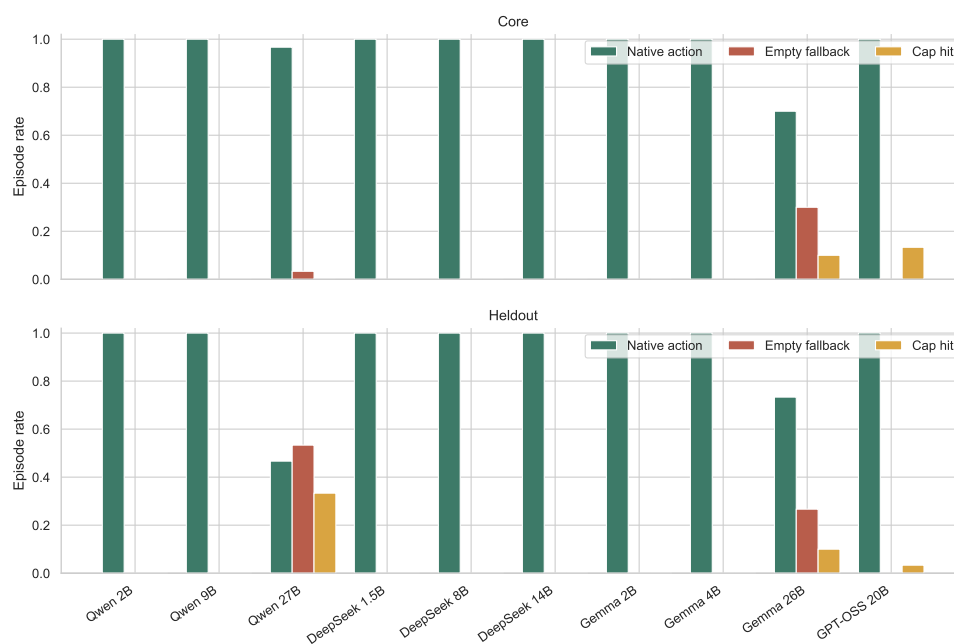


Figure 1: Primary interface diagnostics. Top: observed score versus native-only score. Bottom: native-action, empty-fallback, and cap-hit rates on the frozen primary-task aggregate.

Silent objective shifts remain materially harder than no-shift episodes

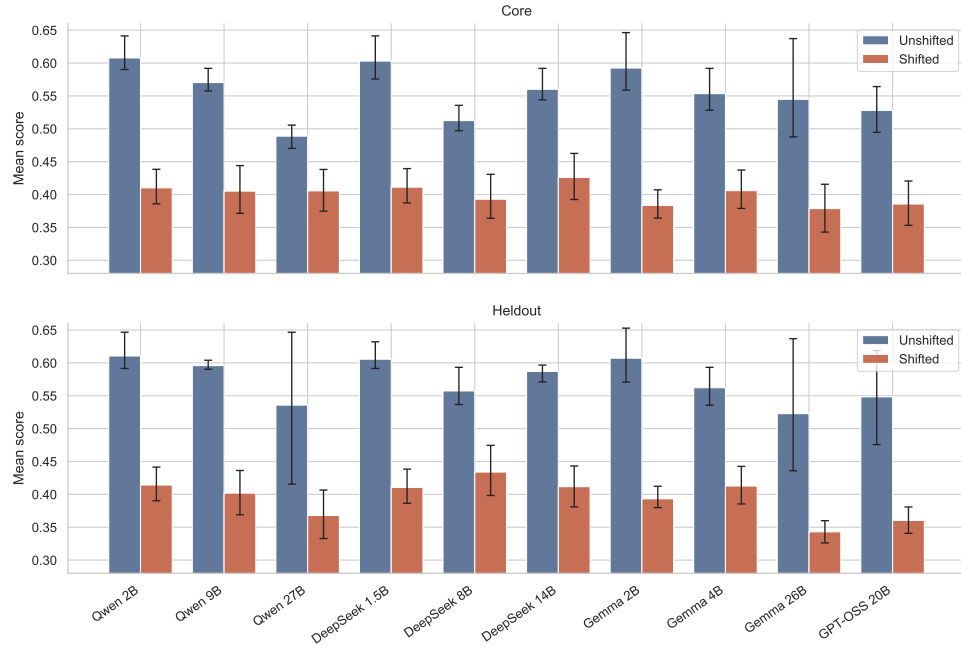


Figure 2: Shifted episodes remain materially harder than no-shift episodes on the frozen primary-task aggregate. Error bars denote 95% bootstrap confidence intervals for the score aggregates.

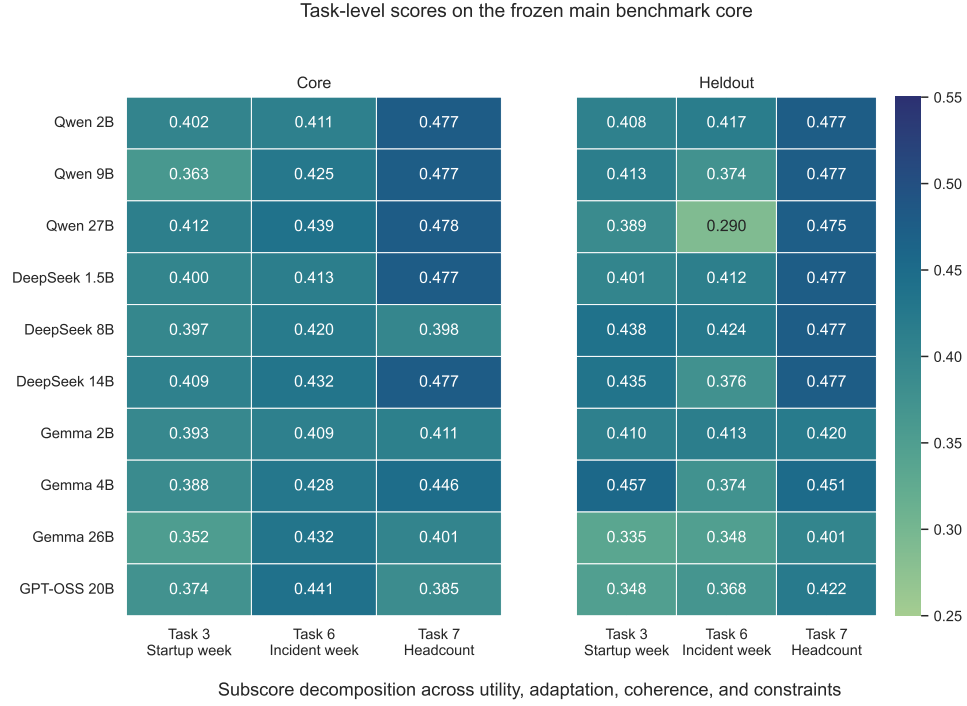


Figure 3: Primary-task heatmaps. Top: task-level performance on the frozen main-task core. Bottom: subscore decomposition across utility, adaptation, coherence, and constraints.

Shell-controlled appendix diagnostic: lightweight parse repair only partially rescues Qwen-27B and does not rescue Gemma-26B

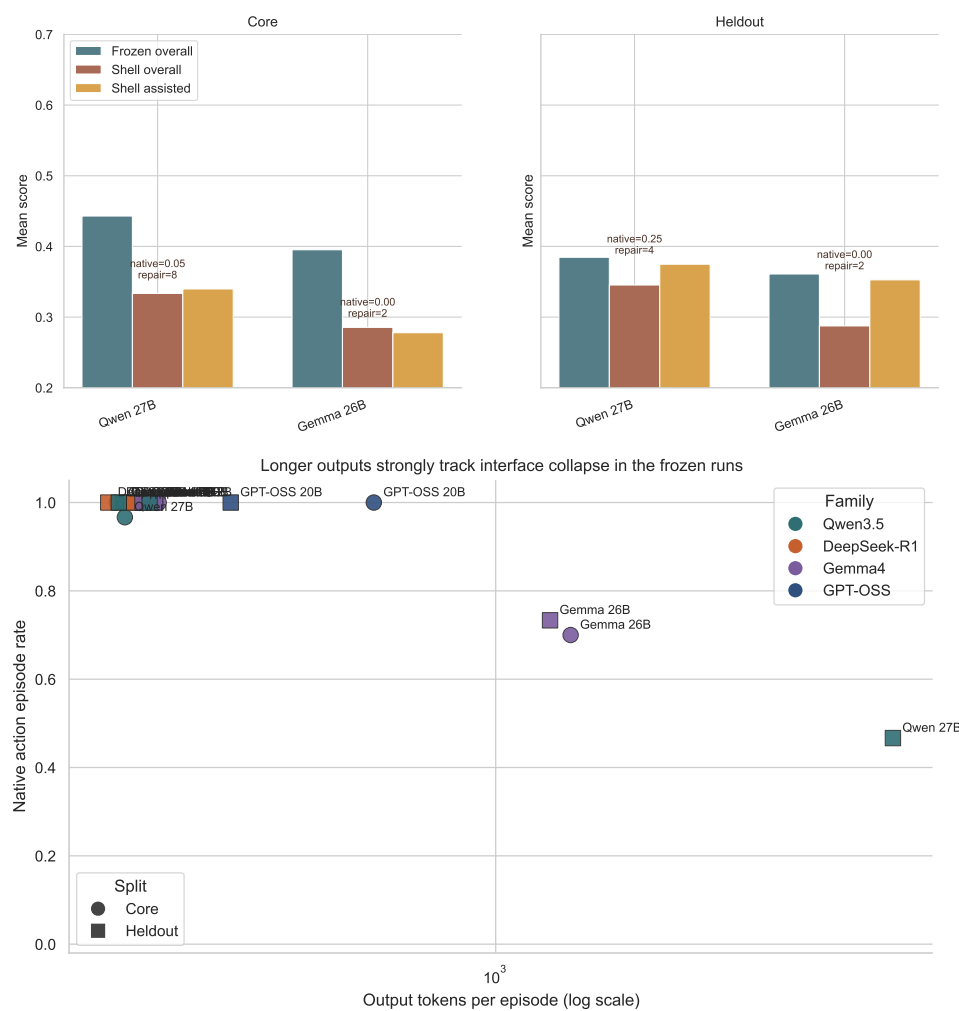
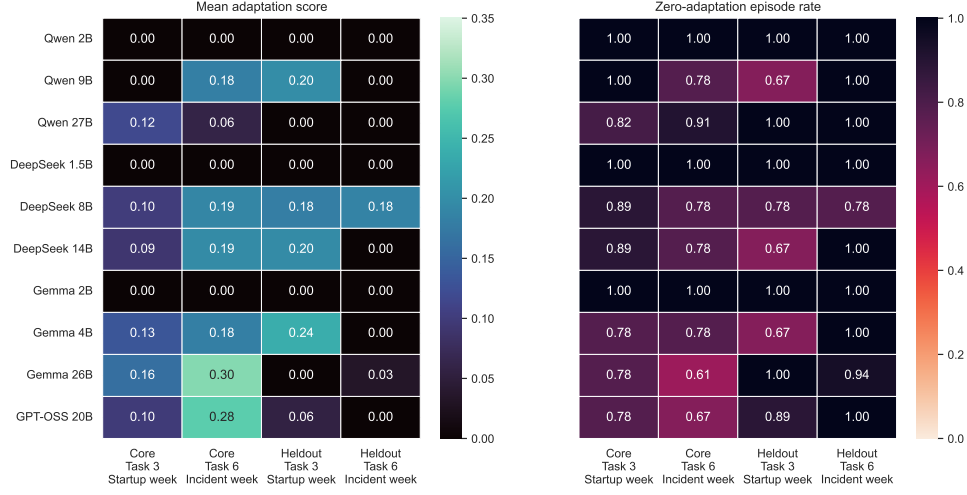


Figure 4: Interface-mechanism diagnostics. Top: shell-controlled diagnostic; parse repair produces only modest improvement for Qwen 27B and does not rescue Gemma 26B. Bottom: token pressure versus native-action reliability; the most shell-fragile large-model runs are also the most verbose.

Adaptation failures are concentrated in specific task/split slices rather than uniformly across the benchmark



Restricted-view leakage probes stay near 4-way chance on the primary tasks

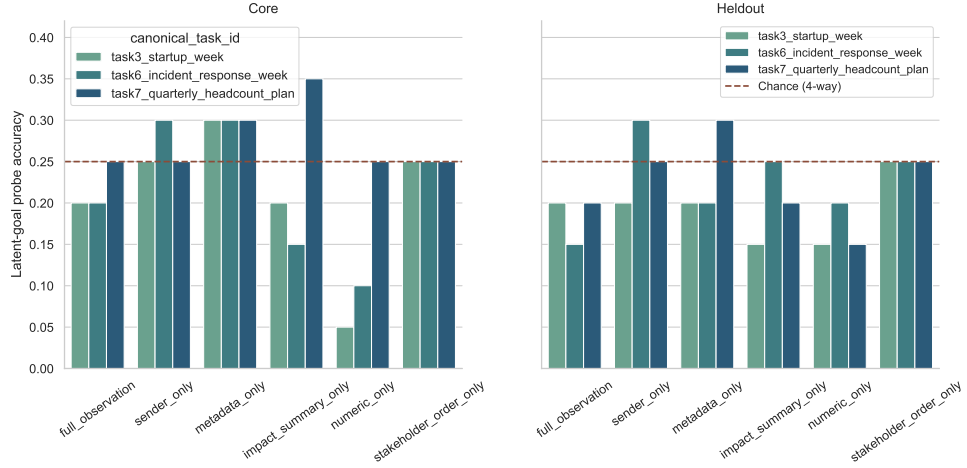


Figure 5: Adaptation and leakage diagnostics. Top: adaptation audit showing mean adaptation score and zero-adaptation episode rate by task and split. Bottom: leakage audit across restricted observation views on the primary aggregate tasks. Chance is 0.25 for the four-way archetype probe, and the explicit-goal-token rate is 0.0 throughout the frozen audit.

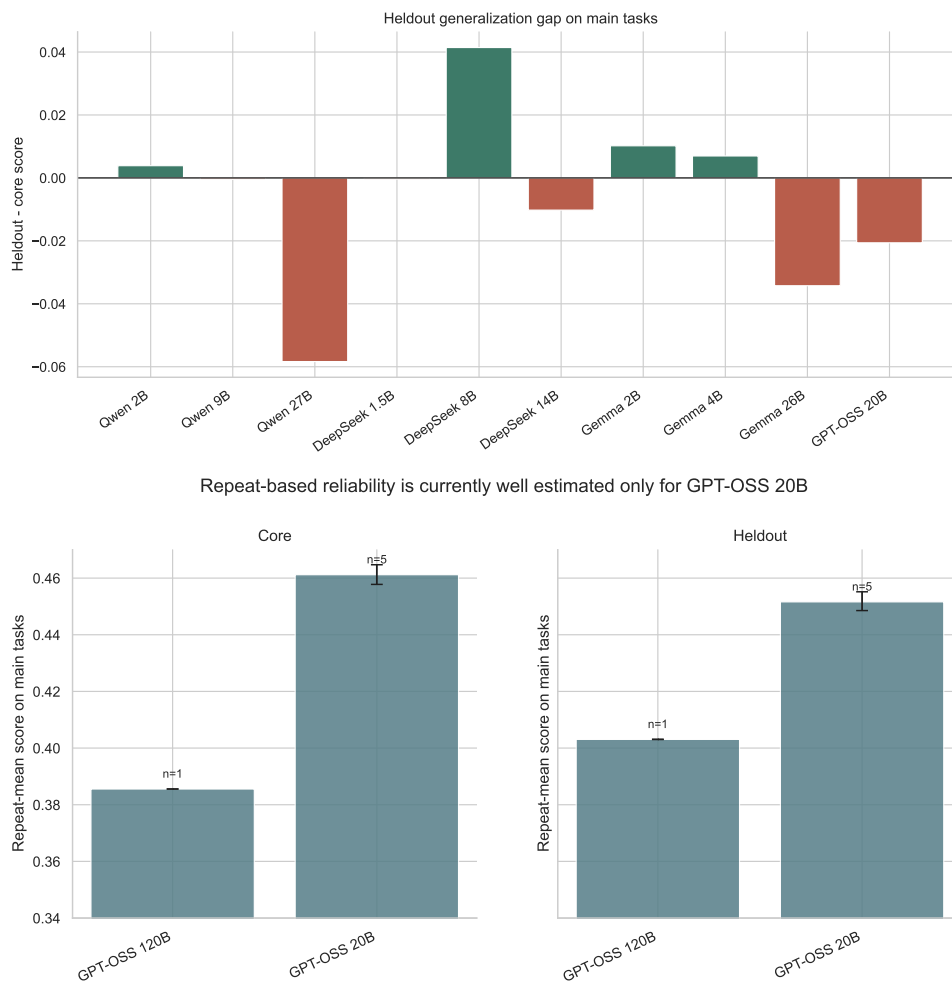


Figure 6: Generalization and repeatability diagnostics. Top: heldout-minus-core gap on the primary aggregate tasks. Bottom: repeat-based reliability estimates where repeat coverage exists; GPT-OSS 120B remains a low-coverage appendix anchor rather than a load-bearing comparison.

122 8 Case Studies and Provenance

123 The frozen bundle includes four qualitative trajectory analyses:

- 124 • `deepseek14_success.md`
- 125 • `qwen27_interface_failure.md`
- 126 • `gemma26_interface_failure.md`
- 127 • `qwen9_native_policy_regression.md`

Model	Split	Task	Score	Dominant diagnosis
DeepSeek 14B	core	Incident Response Week	0.6353	Fully native, coherent incident-focused policy that keeps support and messaging aligned while delayed effects land.
Qwen 27B	core	Incident Response Week	0.2564	Starts with a plausible action, then falls into repeated empty fallbacks after the hidden shift.
Gemma 26B	heldout	Startup Week	0.2561	Mixed shell/policy failure with repeated empty steps, cap hits, and almost no recovery traction.
Qwen 9B	core	Startup Week	0.3070	Fully native trajectory that stays schema-correct but keeps reverting to stale startup moves after the hidden shift.

Table 15: Curated case-study summary used in the supplement discussion.

128 Table 16 adds transcript-style trajectory snippets. Unlike a dialogue benchmark, LatentGoalOps
 129 exposes structured operating state and structured actions, so each row pairs a compact state snapshot
 130 with a paraphrase of the executed action rather than dumping raw JSON payloads.

131 Across the four frozen case studies, similar scalar scores can arise from meaningfully different
 132 mechanisms. DeepSeek 14B provides the clean positive control: the action shell remains native
 133 and the policy stays incident-focused while delayed effects resolve. Qwen 27B shows a different
 134 pattern, where one mid-episode shell break is followed by a policy over-correction into aggressive
 135 startup moves. Qwen 9B is useful because it removes shell collapse from the story almost entirely:
 136 the trajectory stays schema-correct but keeps reverting to stale growth behavior after the hidden
 137 shift. Gemma 26B combines both failure classes, with repeated empty actions and almost no policy
 138 recovery. That qualitative separation is why the paper reports strict score, native-only score, and
 139 native-episode rate together rather than collapsing everything into a single scalar.

140 The strengthened bundle is organized to support direct regeneration of the paper assets:

```

141 outputs/paper-freeze-2026-04-18/
142   tables/
143   figures/
144   case_studies/
145   metadata/

```

146 The `tables/` and `figures/` directories contain the analysis outputs consumed by the manuscript.
 147 The `case_studies/` directory stores curated trajectory narratives, and `metadata/` contains the
 148 frozen manifest and protocol descriptors. The exact upstream raw run directories needed to rebuild
 149 those assets are enumerated in `metadata/run_manifest.json` and are shipped in the anonymous
 150 reviewer archive under their original relative paths rather than duplicated inside the 2026-04-18
 151 analysis root itself.

Step/day	Observed state	Executed action (paraphrased)	What the snippet shows
DeepSeek 14B, core Incident Response Week, score 0.6353, fully native			
Step 0 / Day 2	3 pending effects; early incident week state still unstable	Cut usage pricing 10%, keep retention messaging, keep incident-swarm support, push usage-pricing initiative	Starts with a service-preserving move even before delayed effects land; no shell instability and no thrashing across goals.
Step 2 / Day 4	4 pending effects; incident backlog still active	Optimize infra while keeping retention messaging and incident-swarm support	Mid-episode policy stays incident-focused instead of pivoting to generic growth or cost moves.
Step 5 / Day 7	2 pending effects; delayed effects still resolving	Refactor infra, keep retention messaging, keep incident-swarm support, cut usage pricing 10%	Strong positive reward arrives without the model abandoning its earlier operational stance.
Step 6 / Day 8	0 pending effects; week closes cleanly	Refactor incident tooling, keep retention messaging, keep incident-swarm support	Coherent finish: action shell stays native, and policy remains aligned through episode end.
Qwen 27B, core Startup Week, score 0.3685, shell collapse after hidden shift			
Step 0 / Day 2	5 pending effects; early startup state	Optimize infra, refactor incident tooling, automate support triage; cost-oriented messaging; automation-first support	Plausible pre-shift startup move; qualitative issue is not immediate nonsense but what happens after state changes.
Step 3 / Day 5	0 pending effects; post-shift transition point	Empty fallback action {} after parse failure	Critical shell break: world advances, but model contributes nothing useful at exactly the point where adaptation pressure matters most.
Step 6 / Day 8	5 pending effects; recovery attempt under new state	Ship usage pricing, switch to revenue upsell, move to premium-SLA support	Over-corrects into aggressive revenue posture after the shell break instead of recovering a stable startup policy.
Steps 7–9 / Days 9–11	Pending effects shrink, but score remains negative	Repeats referral-loop plus growth-push moves despite negative rewards	Failure mode is not only one malformed step; policy stays stale and misaligned even after shell recovers.

Table 16: Transcript-style trajectory excerpts paraphrased from the frozen case-study markdowns. These are qualitative observation-action snapshots, not chain-of-thought traces. They make two distinct failure modes concrete: coherent native adaptation versus interface collapse followed by policy over-correction.

9 Reproducibility

The paper assets are generated with:

```
uv run paper-assets --config configs/paper_protocol_strengthened.yaml
```

This command rebuilds the paper tables, figures, and case studies from the strengthened artifact bundle. The manuscript itself can be compiled with the bundled `tectonic` binary included in the repository tools directory. The submission’s anonymous review archive contains the code, the outputs/paper-freeze-2026-04-18 analysis root, the exact raw run directories referenced by the manifest, and the release documents `paper/release/REVIEW_BUNDLE.md`, `paper/release/COMPUTE.md`, and `paper/release/THIRD_PARTY_ASSETS.md`. Those materials make the benchmark runnable for reviewers while keeping the main paper focused on the benchmark claims rather than on packaging detail.

10 Threats to Validity

10.1 Provider heterogeneity

Most frozen runs are local Ollama runs, while GPT-OSS-120B is served through OpenRouter. We therefore treat GPT-OSS-120B as an external anchor and not as a clean within-family local scaling point.

168 10.2 Interface confounding

169 Some large-model drops are clearly interface-mediated. Rather than hiding that fact, the paper
170 makes it part of the result and reports strict overall score, native-only score, fallback rates, and the
171 shell-controlled diagnostic.

172 10.3 Adaptation metric sharpness

173 The adaptation score for P1 and P2 uses a thresholded behavioral detection rule. That choice is
174 operationally meaningful, but it also makes some slices of the benchmark harsher than a fully smooth
175 scalar metric would. The supplement therefore includes both the formal definition and a dedicated
176 adaptation audit.

177 10.4 Synthetic world construction

178 LatentGoalOps is synthetic by design. The claim is not that it replicates every real operating en-
179 vironment; it is that it isolates a capability that existing benchmarks often blur: hidden-objective
180 tracking with delayed effects and silent non-stationarity. The construct-validity argument therefore
181 relies on heldout splits, leakage audits, task design transparency, and baseline separation rather than
182 on realism language alone.

183 References

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